

Simulation for Planetary Robotic Perception and Autonomy: A Concise Survey of Recent Capabilities and Gaps

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Abstract—Planetary robotics is an important enabler of scientific exploration in environments where direct human-in-the-loop operation is costly, hazardous, or infeasible. However, developing and validating planetary robotic systems remains difficult because representative field testing is expensive, limited, and often unrepeatable under mission-relevant conditions. In this setting, simulation serves as a central tool for perception and autonomy research, synthetic data generation, system integration, and pre-deployment evaluation. Despite its importance, the literature on planetary robotics simulation remains dispersed across different simulation engines, implementations, and application settings. This paper surveys simulation works for planetary robotic perception and autonomy across four practical axes: Openness and Availability, Scenario and Platform Coverage, Sensor and Perception Support, and Environmental and Operational Realism. The surveyed simulation works report visual or physical fidelity and support perception-oriented workflows. They also indicate uneven public availability, rover-centered coverage, partial support for specialized sensing modalities, and limited treatment of operational constraints such as on-board computation, energy, and communication restrictions. Detailed selection criteria, the survey database, and resource links with verification dates are available on our project website at <https://kimhoyun-robotair.github.io/Survey-For-Planetary-Robotic-Simulation/>.

I. INTRODUCTION

Space robotics covers robotic systems designed to operate beyond Earth. Prior reviews distinguish orbital and planetary applications as different autonomy contexts [1]: orbital robots typically operate in microgravity around spacecraft, whereas planetary robots must traverse natural surfaces under uncertain terrain, variable illumination, limited communication, and long-duration operational constraints [1], [2]. This paper focuses on the latter class of systems, namely robots intended for planetary surface or near-surface exploration. That class is also broadening beyond wheeled rovers to include legged, aerial, and heterogeneous robotic systems [3], [4].

Simulation is central in this setting because representative validation is expensive, limited, and difficult to repeat. Planetary autonomy also couples perception, localization, motion generation, and decision-making [2]; recent work on perception-aware planning, global localization, and multi-robot coordination further illustrates this coupling [4]–[7]. As a result, simulation works increasingly support more than visualization: they are used for synthetic data generation, perception-oriented evaluation, and software-in-the-loop ex-

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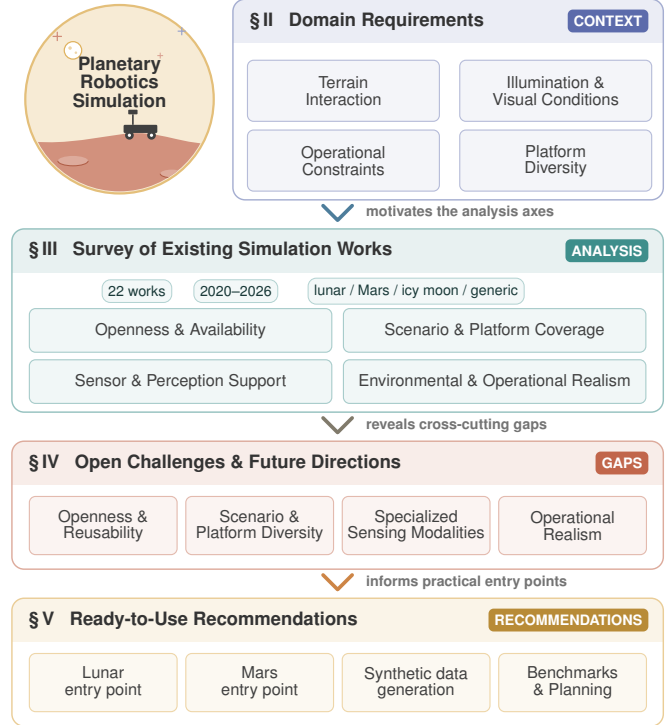


Fig. 1: Structure of this survey. The paper first derives planetary simulation requirements, then surveys twenty-two simulation works along four analysis axes, summarizes the resulting gaps as open challenges, and provides practical starting points for new users.

perimentation [8], [9].

Existing reviews cover important parts of the broader field, including space robotics [1], decisional autonomy [2], active SLAM [30], and AI-enabled vision for space missions [31]. However, they primarily organize the literature around algorithms or missions rather than simulation works. They therefore offer limited guidance on how to choose, compare, or reuse simulation works for perception and autonomy studies. This is a practical gap because simulator choice depends not only on the target algorithm or mission, but also on sensing modalities, robotic platforms, synthetic data needs, environmental conditions, and autonomy-stack compatibility. To address this gap, we survey simulation works as they are used in planetary perception and autonomy research. In this paper, simulation works include simulators, rendering pipelines, synthetic-data generators, and benchmark-oriented environments when they support planetary perception or autonomy research.

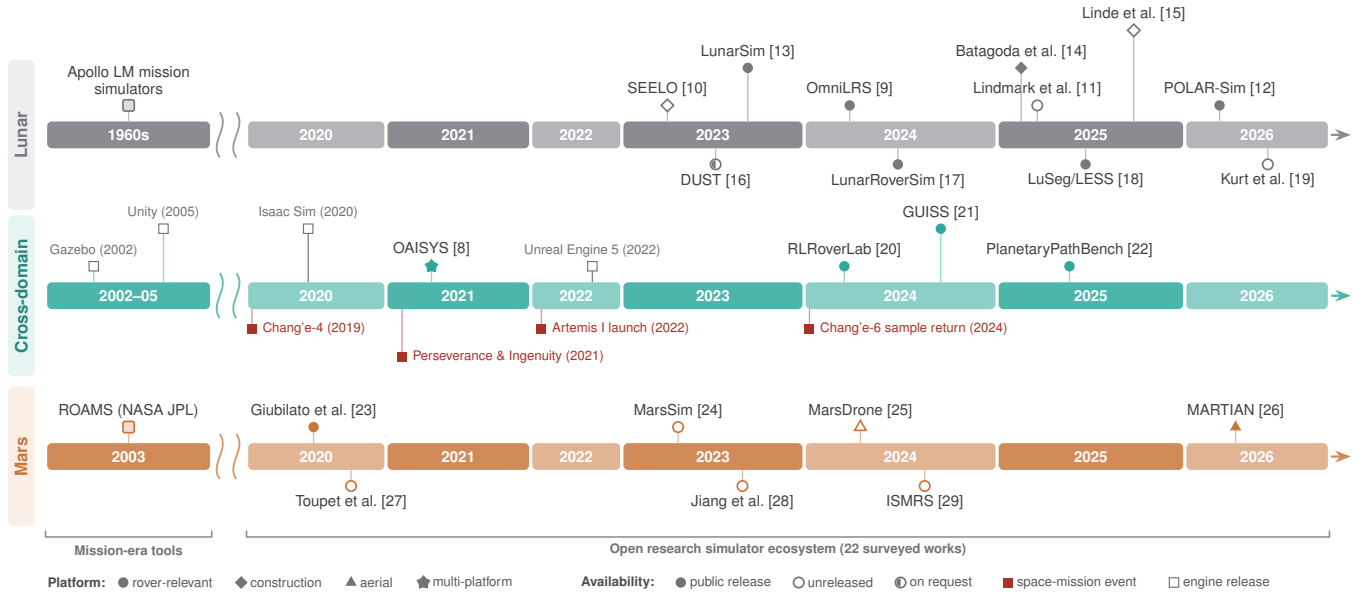


Fig. 2: Chronological roadmap of the surveyed planetary simulation works as domain lanes (top: lunar; middle: cross-domain or other bodies; bottom: Mars). Each lane starts from an early mission-era anchor (Apollo LM mission simulators, the ROAMS rover simulator at NASA JPL, and the initial releases of general-purpose engines; shown as uncited historical markers), and the wavy break compresses the intervening decades before the recent time series. Marker shape encodes the platform or task category, and marker fill encodes public availability (filled: direct release; outline: unreleased; half: on request). Dark-red squares mark major space-mission events and open gray squares mark simulation-engine releases.

This paper addresses that need with three contributions:

- 1) We provide a simulation-centered perspective on recent works for planetary robotic perception and autonomy, organized around four practical axes: openness, scenario coverage, sensing support, and operational realism.
- 2) We relate these axes to planetary autonomy requirements and summarize the reported capabilities of twenty-two surveyed works across access, platform scope, sensing, task outputs, and environmental or operational assumptions.
- 3) We identify recurring gaps in this mapping and discuss future directions toward reusable planetary simulation ecosystems with explicit assumptions.

The remainder of this paper is organized as follows. Section II introduces the mission context and derives the simulation requirements that motivate the survey axes. Section III surveys existing simulation works from four analytical perspectives: Openness and Availability, Scenario and Platform Coverage, Sensor and Perception Support, and Environmental and Operational Realism. Section IV discusses open challenges and future directions, Section V distills practical simulator recommendations for newcomers, and Section VI concludes the paper. Fig. 1 summarizes this structure, and Fig. 2 presents the twenty-two surveyed simulation works chronologically by target domain.

II. PLANETARY ROBOTICS PRELIMINARIES

A. Planetary Robotics as an Integrated Autonomy Problem

Planetary robotics concerns robotic operation on planetary surfaces. In this setting, systems must handle uncertain ter-

rain interaction, including slip, sinkage, and unknown surface properties, as well as significant communication delays with Earth and GNSS-denied localization. They must also operate under extreme environmental constraints, including vacuum, radiation, and wide thermal cycling, during missions that can span months to years with minimal human-in-the-loop operation. Early work already treated perception, localization, motion generation, temporal planning, and resource management as components of one operational system [2], and later studies on terrain-adaptive navigation, global localization, and perception-aware planning reinforce this coupling [5]–[7], [32]–[34].

These constraints distinguish planetary robotics from terrestrial autonomy in ways that directly affect simulation design. Fig. 3 summarizes the main domain-specific factors and their simulation implications.

B. Implications for Simulation

A planetary robotics simulator should support the task-relevant physical, sensing, and operational assumptions behind a study, not only render planetary scenes. These domain factors motivate four considerations for perception-oriented simulation. First, the simulator should represent terrain and robot–environment interaction at the level required by the target task [35]–[37]. Second, it should reproduce sensing conditions that matter for localization or scene interpretation, including low texture and extreme illumination [5], [6], [38], [39]. Third, it should make the operational assumptions of the target autonomy stack explicit, including software integration, communication delay, onboard computation, and energy budgets [2], [40], [41]. Fourth, it should account for

Domain Factor	Planetary Characteristic	Simulation Implication (This Survey)
PHYSICAL ENVIRONMENT		
■ Terrain interaction	Slip, sinkage, and wheel–regolith coupling	Physics-based terramechanics
■ Illumination and visual conditions	Low texture and extreme illumination	Physically-based rendering and sensor modeling
■ Gravity and contact regime	Low-gravity regimes	Configurable gravity and contact dynamics
OPERATIONAL CONSTRAINTS		
■ Communication	Communication delay and limited bandwidth	Latency-aware autonomy modeling
■ Onboard resources	Bounded power, compute, and mission duration	Energy- and compute-aware evaluation
ROBOTIC SYSTEM SCOPE		
■ Platform diversity	Rovers, legged robots, aerial vehicles, and multi-robot systems	Heterogeneous platform support
■ Sensor and autonomy stack	Integrated perception, localization, planning, and resource management	Task-relevant sensing and stack integration

Fig. 3: Key factors distinguishing planetary from terrestrial autonomy and their implications for simulation design. Each domain factor (left) is linked to its reported planetary characteristic (center) and the simulation capability it implies (right), grouped into physical environment, operational constraints, and robotic system scope.

the diversity of planetary robotic platforms, including rovers, legged robots, aerial vehicles, and multi-robot systems [3], [42], [43].

These considerations motivate the survey axes in Section III. Because representative planetary testing is difficult to reproduce, we also treat openness and availability as a practical axis for reuse and comparison, as illustrated in Fig. 1.

III. SURVEY OF EXISTING SIMULATION WORKS

A. Taxonomy and analysis criteria

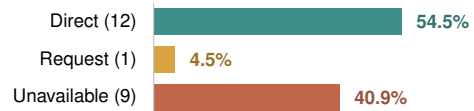
Building on Section II, we organize recent planetary simulation works along four analysis axes: openness and availability, scenario and platform coverage, sensor and perception support, and environmental and operational realism. The surveyed set is intentionally bounded in scope: we focus on works from 2020 onward with publicly accessible scholarly documentation that explicitly present planetary simulation, rendering, synthetic-data generation, or benchmark works relevant to robotic perception and autonomy. We therefore exclude general-purpose engines by themselves, internal tools, challenge-only environments, and projects without accompanying scholarly documentation.

We analyze these entries as a heterogeneous set of works, not as directly comparable complete simulators. Here, rover-relevant denotes works that are not full rover simulators but still provide terrain, stereo/depth, planning, or perception inputs useful for rover autonomy studies. Table I summarizes the resulting twenty-two works, and the following distributions should be read as descriptive patterns within this set, not as field-wide prevalence estimates.

B. Openness and Availability

Within the surveyed set, twelve of twenty-two works provide direct public repositories or datasets, one provides

(a) Online Availability



(b) Platform Category

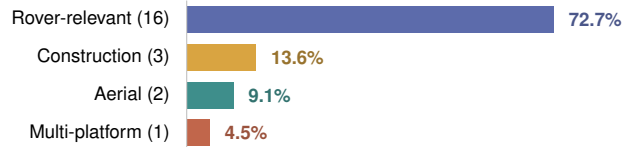


Fig. 4: Distribution of surveyed simulation works by (a) public availability and (b) primary platform or task relevance.

access through a request/download channel, and nine are not accompanied by a public release (Fig. 4a). Publicly available examples include the framework of Giubilato et al. [23], OAISYS [8], OmniLRS [9], LunarSim [13], RL-RoverLab [20], LunarRoverSim [17], Batagoda et al. [14], LuSeg/LESS [18], GUISS [21], PlanetaryPathBench [22], MARTIAN [26], and POLAR-Sim [12]. DUST is separately marked as partially available because its release is provided through a NASA request/download channel, not as a directly hosted repository [16]. The remaining nine works are described without a corresponding online release.

This distribution highlights that access to the underlying engine and access to the actual study artifact are separate issues: a work may build on a common engine while its implementation, dataset, or benchmark remains unavailable.

C. Scenario and Platform Coverage

The surveyed works remain centered on rover scenarios and rover-relevant terrain or planning tasks (Fig. 4b), consistent with the historical role of rovers in planetary

TABLE I: Overview of the twenty-two surveyed simulation works: target domain, primary platform category, engine or tool basis, middleware support, access status, terrain data source, and supported sensors or outputs. Access denotes the reported work release when available; for engine-based works, the linked repository may correspond to the released implementation or its public engine basis. St = Stereo; Sem = Semantic; Seg = Segmentation; Enc = Wheel encoder; Alt = Altimeter; Inc = Inclinometer.

Work	Year	Domain	Category	Engine / Tool	ROS	Access	Terrain Source	Sensors / Outputs
Giubilato et al. [23]	2020	Mars	Rover	Gazebo	R1	Repo	Real	St Cam, 3D LiDAR, IMU
Toupet et al. [27]	2020	Mars	Rover	ROS-based	R1	–	Simulated	St Cam, rover state
OAISYS [8]	2021	Planet.	Multi-platform	Blender	–	Repo	Synthetic	RGB-D (extensible)
MarsSim [24]	2023	Mars	Rover	Gazebo	R1	–	Real	Mastcam
SEELO [10]	2023	Lunar	Excavator	Unity	–	–	Real	Vision, IMU
LunarSim [13]	2023	Lunar	Rover	Unity	R2	Repo	Synthetic	RGB-D, IMU
Jiang et al. [28]	2023	Mars	Rover	Blender	–	–	Synthetic	RGB-D + Sem
DUST [16]	2023	Lunar	Rover-relevant	Unreal Engine	–	Request	Real/DEM	Rendered terrain views
OmniLRS [9]	2024	Lunar	Rover	Isaac Sim	R1/R2	Repo	Real + proc.	RGB-D, 2D/3D LiDAR, IMU
MarsDrone [25]	2024	Mars	Aerial	Unreal Engine	–	–	Real	Cam, Alt, Inc, IMU
RLRoverLab [20]	2024	Lu./Ma.	Rover	Isaac Sim	–	Repo	Synthetic	RGB Cam
LunarRoverSim [17]	2024	Lunar	Rover	PyBullet	–	Repo	Config. lunar	RGB-D + Seg
ISMRS [29]	2024	Mars	Rover	Isaac Sim	R2	–	Real	St Cam, IMU, LiDAR, Enc
GUISS [21]	2024	Icy moon	Rover-relevant	Blender-based	–	Repo	Synthetic	Stereo imagery
Batagoda et al. [14]	2025	Lunar	Excavator	Chrono	R2	Repo	Real (Earth)	Cam, sensor sim.
LuSeg/LESS [18]	2025	Lunar	Rover	UE + AirSim	R1/R2	Repo	–	RGB-D + Seg
Linde et al. [15]	2025	Lunar	Exc./Truck	AGX Dynamics	R2	–	Synthetic	Observation sensor
Lindmark et al. [11]	2025	Lunar	Rover	OpenPLX + AGX	R2	–	LDEM-derived	RGB-D, LiDAR, IMU
PlanetaryPathBench [22]	2025	Ma./Lu.	Rover-relevant	Benchmark maps	–	Repo	Real/DEM	Occupancy maps
MARTIAN [26]	2026	Mars	Aerial	Blender	–	Repo	Real map	Aerial RGB + pose
POLAR-Sim [12]	2026	Lunar	Rover	Chrono	–	Repo	Real + synth.	RGB, annotations, terrain meshes
Kurt et al. [19]	2026	Lunar	Rover	MuJoCo + Isaac	–	–	Synthetic	RGB-D + pose

surface exploration. Toupet et al. [27] and POLAR-Sim [12] further reinforce this rover-oriented emphasis through Mars 2020 autonomous navigation testing and lunar perception-oriented rover simulation. At the same time, OmniLRS reports quantitative photorealism evaluation for lunar-rover simulation [9], SEELO [10] and Batagoda et al. [14] address lunar excavation and construction-oriented simulation, MarsDrone [25] considers aerial platforms, and MARTIAN [26] targets Mars aerial navigation imagery for map-based localization. The surveyed works also include task-specific simulation efforts: DUST [16] for lunar terrain and illumination analysis, GUISS [21] for icy-moon stereo/depth simulation, PlanetaryPathBench [22] for terrain-derived rover path-planning benchmarks. Even so, heterogeneous and mission-scale scenarios are still treated mainly as specific extensions, with few works making them default assumptions.

D. Sensor and Perception Support

Sensor and perception support is one of the clearest convergence points across the surveyed literature. Representative examples include multi-modal synthetic data generation, computer-vision prototyping, segmentation-oriented lunar datasets, map-based localization imagery, and vision-oriented rover navigation [9], [13], [18], [19], [26].

Table II contrasts the sensing modalities emphasized in current simulation works against those discussed in planetary mission documentation and navigation surveys. The dominant supported modalities are cameras, RGB-D, LiDAR, and IMUs, together with perception-oriented outputs such as semantic labels, depth, and annotations. However, explicit support for more specialized modalities (ground-penetrating radar, thermal, event-based, celestial-navigation sensors) is reported much less often. Table I further details the terrain

TABLE II: Sensing modality coverage reported in the surveyed works and in mission, planned-mission, prototype-stage, or agency research-stage literature. Entries indicate reported support or use rather than comprehensive coverage of the field or baseline simulator requirements.

Sensing Modality	Surveyed Works	Mission / Prototype
Mono / Stereo Camera	✓	✓
RGB-D	✓	✓
2D / 3D LiDAR	✓	✓
IMU	✓	✓
Wheel Encoder / Odometry	○	✓
Semantic / Instance Labels	✓	–
Star Tracker / Celestial Nav.	✗	✓ [44]
Ground-Penetrating Radar	✗	✓ [45], [46], [47]
Thermal / IR Sensor	✗	✓ [48], [49]
Event-based Camera	✗	○ [50]
ToF / Flash LiDAR	✗	○ [51], [52]

✓ = reported support or use; ✗ = not reported in the reviewed descriptions; ○ = limited or indirect support; entries are based on how each work reports its capabilities.

Reported support denotes capabilities implemented and exposed by the work itself; engine-level possibilities that require substantial user-side implementation are not counted.

data source and the supported sensor modalities and outputs per work.

E. Environmental and Operational Realism

Environmental and operational realism are treated here as complementary aspects of planetary simulation. The former includes terrain geometry, regolith contact, illumination, gravity, and sensing conditions, while the latter concerns communication delay, onboard computation, energy use, and autonomy-stack execution. Fig. 5 summarizes where these factors appear in a mission scene and how they are

represented across the surveyed works.

Within the surveyed set, recent works report terrain representation or deformation [10], [12], [16], [22], physics-based sensor simulation [12], [14], and mission-oriented workflow analysis including energy consumption or rover navigation testing [15], [27]. Related recent studies outside the core set further examine terrain deformation and regolith interaction for lunar rover simulation [53], [54]. Other task-focused works contribute stereo/depth-oriented surface simulation for icy-moon autonomy [21], controlled Mars aerial image rendering from orbital map products [26], and coupled contact-dynamics and vision-oriented rover simulation [19]. Across these works, environmental assumptions are often reported through terrain, contact, illumination, or rendering choices, whereas operational constraints such as communication delay, onboard computation, and energy budgets are reported less consistently.

IV. OPEN CHALLENGES AND FUTURE DIRECTIONS

The four analysis axes of Section III reveal corresponding open challenges, as traced in Fig. 1. This section discusses these challenges and concludes with a unifying direction.

A. Limited openness and reusable public ecosystems

As summarized in Section III-B, public availability remains uneven across the surveyed works. For planetary robotics, this is especially relevant because representative field testing and real mission data are limited, expensive, and difficult to reproduce. When a simulation work is publicly accessible, later studies can reuse its assets, compare methods under shared settings, and reduce repeated engineering. A practical future direction is broader release of simulation software, assets, scenes, sensor configurations, and evaluation components whenever project constraints permit.

B. Rover concentration and limited scenario diversity

The surveyed literature remains rover-centered, but the broader planetary robotics mission space already includes cooperative exploration [3], [4], aerial vehicles [42], and multi-robot operation [43]. NASA reporting has shown that Ingenuity provided scouting support for Perseverance, illustrating the relevance of heterogeneous aerial-ground cooperation [55]. Similarly, CADRE is designed around a team of small lunar rovers that conduct autonomous distributed measurements, including surface imaging and subsurface mapping with ground-penetrating radar [46], while CRES-CENT addresses constrained lunar rover planning for the same multi-agent mission [56]. These examples show that the gap is broader than platform coverage alone. Reusable scenarios remain limited for heterogeneous, cooperative, and science-driven operations such as planetary skylight and lava-cave exploration [4]. Future planetary simulation work would therefore benefit from broader scenario coverage aligned with such mission concepts and recent fielded or planned robotic systems.

C. Insufficient support for diverse sensing modalities

As shown in Table II, the sensing assumptions in the surveyed simulation works are narrower than the broader sensing possibilities discussed in planetary navigation and mission documentation. The relevant issue is not whether every simulation work should implement every mission payload, but whether task-relevant sensing assumptions can be studied when they matter for localization, traversability estimation, or autonomy. Ground-penetrating radar illustrates this distinction. RIMFAX on Perseverance and the planned CADRE mission show the mission relevance of subsurface radar [45], [46], while recent work [47] explores GPR as a localization modality through subsurface feature matching, expanding its role from scientific sounding to autonomy-oriented state estimation. Other navigation-oriented modalities point to the same issue. Star trackers and celestial-navigation sensors are discussed in lunar surface navigation surveys [44]. Thermal/IR sensing, event-based vision, and ToF/flash LiDAR provide additional examples from mission, prototype, or agency research-stage contexts [48], [50], [51]. Broader simulation support for these modalities would expand the range of algorithmic studies that can be performed before field deployment, including subsurface-aided localization, thermal traversability assessment, event-based perception, and flash-LiDAR hazard detection. This would make planetary simulation a stronger tool for developing and stress-testing perception and autonomy methods under mission-relevant sensing assumptions.

D. Limited treatment of operational constraints

VIPER documentation [57] reports power, communication, and round-trip delay considerations. The lunar surface navigation survey [44] identifies computational cost and hardware requirements for several navigation modes. Operational constraints also appear at the autonomy-stack level. Toupet et al. [27] present a ROS-based simulator for testing the Mars 2020 enhanced autonomous navigation system. The Mars 2020 Autonomous Rover Navigation paper [58] describes how image processing, navigation processing, and driving are overlapped during a traverse, and notes that without this parallelization the drive rate would be more than halved. Compared with these mission-level and software-level considerations, the current surveyed simulation works report onboard computation limits, communication delay, and energy constraints less consistently than terrain or visual assumptions (Fig. 5). Future simulation studies would be stronger if they made these execution assumptions explicit alongside terrain, visual, and sensing assumptions.

E. Toward perception-oriented planetary simulation ecosystems

Taken together, these challenges point toward planetary simulation works that are reusable, interoperable, and explicit about their assumptions. Such works would better support perception and autonomy studies that combine task-relevant sensing, heterogeneous robotic scenarios, execution constraints, and mission-relevant environmental conditions.

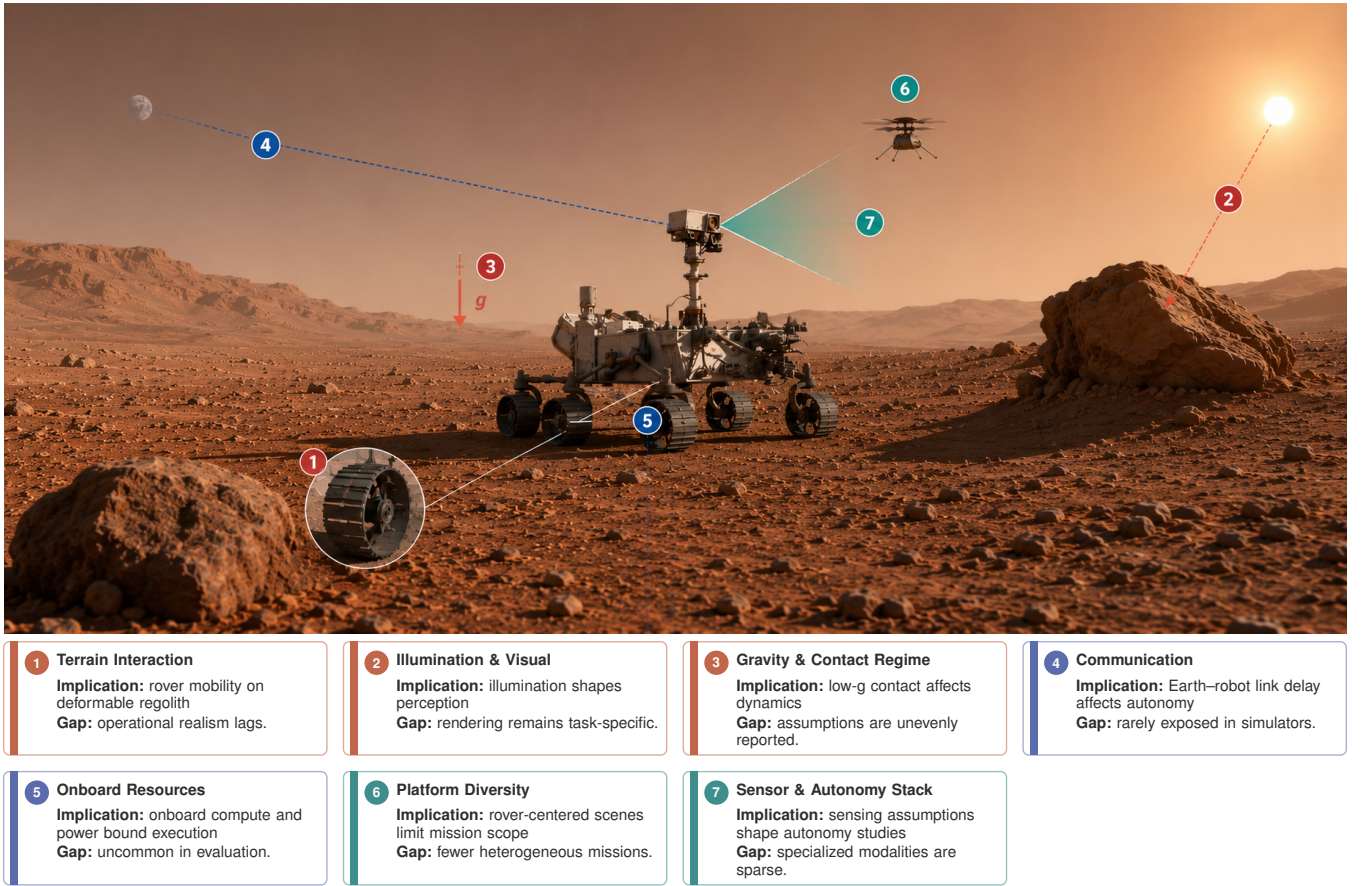


Fig. 5: Mission-scene view of how environmental and operational realism factors appear in planetary simulation. The upper image provides the mission context¹, while the indexed blocks give an at-a-glance summary of the implication and reporting gap associated with each factor. The figure highlights that operational constraints such as communication delay, onboard computation, and energy budgets are reported less consistently than terrain, contact, illumination, or rendering assumptions.

V. READY-TO-USE RECOMMENDATIONS

For researchers entering planetary robotics simulation, the surveyed landscape can be condensed into a small set of practical starting points, selected from works that combine a public release (§ III-B) with documentation and an actively maintained engine ecosystem. For lunar rover studies, OmniLRS [9] is a practical entry point: it couples photorealistic Isaac Sim rendering with ROS 1/ROS 2 interfaces, multi-robot support, and multi-modal synthetic data generation. Lighter-weight computer-vision prototyping is served by LunarSim [13], while RL RoverLab [20] targets learning-based rover autonomy on both lunar and Martian terrain. For Mars rover work in a classic ROS/Gazebo stack, the framework of Giubilato et al. [23] remains a directly reusable baseline. When terramechanics and physical fidelity dominate the research question, Chrono-based works provide relevant starting points, including Batagoda et al. [14] and POLAR-Sim [12], the latter adding lunar lighting realism and a rover digital twin. For synthetic data generation beyond a single platform, OASYS [8] provides a Blender-based pipeline with semantic metadata, and MARTIAN [26] renders Mars

aerial imagery for map-based localization studies. Finally, PlanetaryPathBench [22] offers terrain-derived benchmark maps for evaluating rover path planning without a full simulation stack. These recommendations are intended to help first-time users identify suitable starting points based on their research goals and required simulation capabilities.

VI. CONCLUSION

This paper surveyed twenty-two simulation works for planetary robotic perception and autonomy from four practical perspectives: openness and availability, scenario and platform coverage, sensor and perception support, and environmental and operational realism. The surveyed works show clear progress in photorealistic rendering, synthetic-data generation, annotation support, terrain interaction modeling, and selected physics-based sensor models. At the same time, the analysis identifies recurring gaps: public availability remains uneven, the landscape is still rover-centered, specialized sensing modalities are only partially represented, and operational constraints such as onboard computation, energy, and communication are reported less consistently than visual or terrain assumptions.

These findings suggest that future planetary simulation works should support perception and autonomy as integrated

¹The Mars scene image in Fig. 5 was generated using ChatGPT Image 2.0.

mission-relevant problems. Stronger simulation works should make their assumptions explicit, expose reusable scenes and evaluation components when possible, and cover task-relevant sensing, heterogeneous robotic scenarios, autonomy-stack execution, and mission constraints. The practical recommendations in Section V are intended to help newcomers choose suitable starting points while highlighting where future planetary simulation works can better support reproducible algorithm development.

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